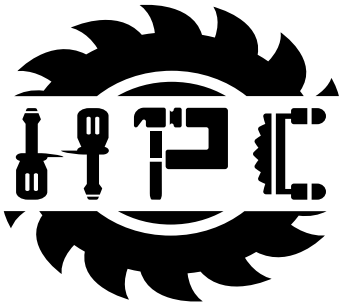


CARPENTRY

HPC Carpentry

A scalable, peer-reviewed training program to democratize HPC access

Northeast Carpentries Community Meeting
May 1, 2026

Our Team:

- Andrew Reid (he/him), US NIST
- Annajiat Alim Rasel (he/him), BRAC U
- Alan O’Cais (he/him), CECAM
- Trevor Keller (they/them)
- Wirawan Purwanto (he/him), ODU

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A scalable, peer-reviewed training program to democratize HPC access

Outline

This is necessarily a brief introduction to our effort

- General overview of the Carpentries
- How HPC Carpentry fits into and builds upon the Carpentries
- Brief history of the HPC Carpentry effort
- Our Strategic Plan
- Challenges

The Carpentries

Motivated by a common problem in research practice – increasingly foundational computational research work was done by a variety of practitioners, often short-timers, with a variety of skill sets, and a variety of practices.

Founded by Greg Wilson starting in the late 1990s, with various structural changes in response to feedback over the years – current umbrella organization (2018) encompasses several sub-organizations – software carpentry, library carpentry, and data carpentry.



Central idea: The skills necessary to demonstrate the value of better practices can be taught in a workshop setting working hands-on with the actual systems – learners build "muscle memory" of having done key steps in the process, and come away with the power to move themselves forward.

- Pedagogically sound – relies on peer-reviewed educational literature
  - Formal instructor training and instructor trainer training
- Open-source – uses instructor and learner feedback to modify lessons
- Inclusive – anyone can contribute feedback, pull-requests



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Common challenges with HPC environment:

- Researchers with HPC requirements often do not have adequate training in operating HPC systems, and struggle to effectively scale up/out their operations even when the hardware resources are available.

- HPC facility operators are frustrated by non-knowledgeable users making inappropriate use of shared resources, disrupting other users, and requiring significant hand-holding before they can become effective.

⇨ Candidate solution: Use Carpentries techniques to build the bridge for HPC users.

Central idea: The skills necessary to demonstrate the value of better practices can be taught in a workshop setting working hands-on with the actual systems – learners build “muscle memory” of having done it, and come away with the power to move themselves forward.

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Some Important Milestones

- ComputeCanada (now Digital Research Alliance of Canada) contributions
- Peter Steinbach, “HPC In A Day”
- SC17 BoF
  - Andy Turner, Christina Koch, Tracy Teal, Bob Freeman, Chris Bording
- CarpentryCon 2018 discussions
- SC18 BoF
  - Andy Turner, Christina Koch, Peter Steinbach, Alan O’Cais, Jeffrey Stafford, John Simpson, Daniel Smith, Bob Freeman
- Well-attended informal session at SC19
- CarpentryCon 2020@Home
  - Trevor Keller, Christina Koch, and others
- SC21 BoF – high-value feedback from HPC operators
- Present in the Carpentries Incubator, Winter 2021
- Begin working towards Lesson Program Incubation
- CarpentryCon 2022
  - Lightning talk, sprint, breakout session
- June 2023 – formal acceptance into Lesson Program Incubation
- June 2024 – LLNL workshop, first road-test of the full workshop
- February 2026 – beginning of the adoption process
- September 2026 – adoption complete

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Strategic Plan

- Build two two-day workshop tracks
  - UNIX Shell*, HPC Intro, and HPC Workflow† lessons, with Admahl's Law application, for HPC users
  - UNIX Shell*, HPC Intro, and HPC Development†† lessons, for coders to learn parallel frameworks
- Identify learner communities and hold workshops
- Reach out to other organizations focused on HPC education
- Continue to curate and host additional material across a spectrum of use cases

Our Current Lessons

- HPC Intro (Queuing system basics) (Beta, in Carpentries Incubator)
- HPC Workflow (Alpha, teachable but preliminary)
- HPC Parallel Novice (Parallelization using Python)
- HPC Shell (deprecated in favor of the Carpentries UNIX Shell lesson)
- HPC Chapel†

- Notes: * Software Carpentry. † Under development. †† Planned.

Our principal audience is novice HPC users. Where possible and practical, we are also interested in serving less-novice HPC users, as well as HPC facility operators. The more community participates, the better we can do together!

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Challenges

- Finding learners
  - We think we address an important problem, but not everyone agrees
  - Many HPC centers have existing training they like
    - But it may not be suitable for beginners?
- Ensuring workshops have clusters to teach on!
  - We have Terraform/Magic Castle tools to stand up clusters
    - But orgs might have to pay for the cloud to do it on
- Making lessons focused
  - We don’t want too many “black boxes”, so it’s tempting to dive in to conversations about MPI, Dask, etc.
  - But the focus of the lesson is cluster ops, not code!

We have a motivated and skilled team, and we feel that these are all fixable with continued outreach and connection to the broader HPC Education world!



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Getting Involved

The GitHub Project Page:
`github.com/hpc-carpentry`

The Slack:
`#hpc-carpentry` on **`carpentries.slack.com`**
`( slack-invite.carpentries.org )`

The Topicbox E-mail List:
`carpentries.topicbox.com/groups/discuss-hpc`

The main website:
`hpc-carpentry.org`



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HPC Carpentry logo is by Mumtahinah Rahman Sristy, BRAC U, Bangladesh (Jan. 2021).